

EHS POC :

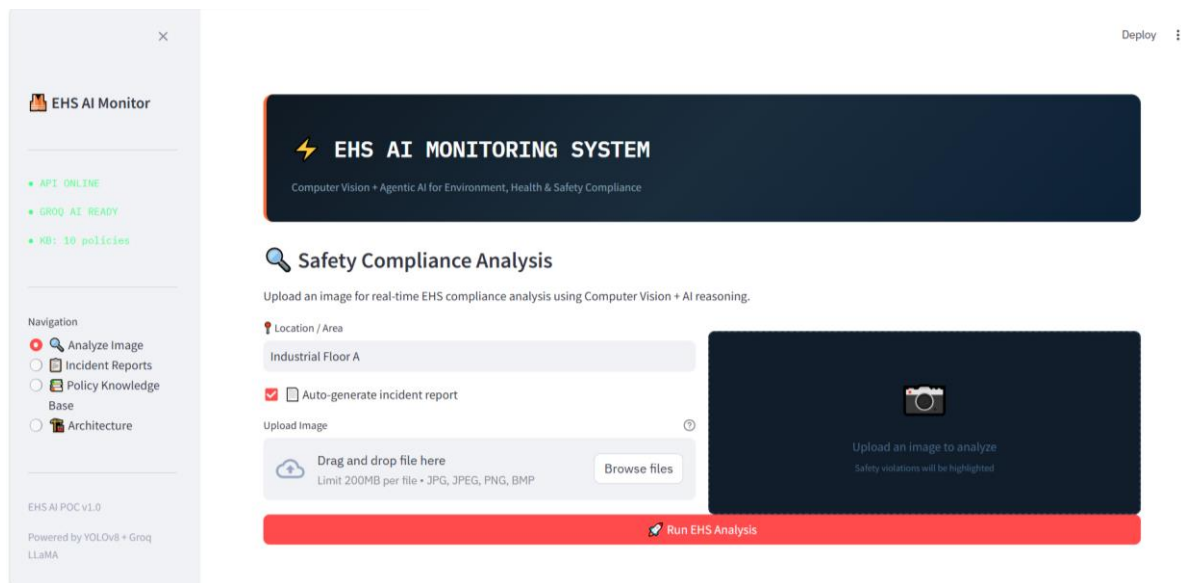
Starting the Server and Data Ingestion:

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←[32mINFO←[0m: Will watch for changes in these directories: ['E:\\Gen AI\\aragen\\ehs_poc\\backend']
←[32mINFO←[0m: Uvicorn running on ←[1mhttp://0.0.0.0:8000←[0m (Press CTRL+C to quit)
←[32mINFO←[0m: Started reloader process [←[36m←[1m19116←[0m] using ←[36m←[1mWatchFiles←[0m]
←[32mINFO←[0m: Started server process [←[36m44912←[0m]
←[32mINFO←[0m: Waiting for application startup.
←[32mINFO←[0m: Application startup complete.
2026-05-18 00:41:39,193 [INFO] watchfiles.main - 2 changes detected
E:\\Gen AI\\aragen\\ehsEnv\\Lib\\site-packages\\requests\\__init__.py:109: RequestsDependencyWarning: urllib3 (2.7.0) or chardet
(7.4.3)/charset_normalizer (3.4.7) doesn't match a supported version!
warnings.warn(
2026-05-18 00:41:39,546 [INFO] watchfiles.main - 5 changes detected
2026-05-18 00:41:39,899 [INFO] watchfiles.main - 1 change detected
2026-05-18 00:41:47,992 [INFO] sentence_transformers.SentenceTransformer - Load pretrained SentenceTransformer: all-MiniLM-L6-v2
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - 🧯 Starting policy ingestion...
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: PPE-001
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: FIRE-001
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: CHEM-001
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: ELEC-001
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: FALL-001
2026-05-18 00:41:52,731 [INFO] knowledge_base_module - Indexing policy: ERGON-001
2026-05-18 00:41:52,735 [INFO] knowledge_base_module - Indexing policy: FORKLIFT-001
2026-05-18 00:41:52,735 [INFO] knowledge_base_module - Indexing policy: WASTE-001
2026-05-18 00:41:52,735 [INFO] knowledge_base_module - Indexing policy: INCIDENT-001
2026-05-18 00:41:52,735 [INFO] knowledge_base_module - Indexing policy: CONFINED-001
Batches: 0% | 0/3 [00:00<?, ?it/s]2026-
05-18 00:41:52,795 [INFO] watchfiles.main - 3 changes detected
Batches:
100% |
| 3/3 [00:00<00:00, 3.34it/s]
2026-05-18 00:41:53,699 [INFO] knowledge_base_module - ✅ Indexed 10 policies (70 chunks) into ChromaDB
2026-05-18 00:41:53,715 [INFO] watchfiles.main - 9 changes detected
2026-05-18 00:41:54,073 [INFO] watchfiles.main - 1 change detected
2026-05-18 00:41:54,491 [INFO] cv_module - Person model loaded
2026-05-18 00:41:54,524 [INFO] cv_module - PPE model loaded
INFO: 127.0.0.1:56274 - "GET /health HTTP/1.1" 200 OK
E:\\Gen AI\\aragen\\ehsEnv\\Lib\\site-packages\\requests\\__init__.py:109: RequestsDependencyWarning: urllib3 (2.7.0) or chardet (7.4.3)/charset_normalizer (3.4.7) doesn't match a supported version!
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05-18 00:41:52,795 [INFO] watchfiles.main - 3 changes detected
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INFO: 127.0.0.1:56274 - "GET /health HTTP/1.1" 200 OK

```

FrontEnd: Streamlit UI:



Running End to End Pipeline:

INFO: 127.0.0.1:56310 - "GET /health HTTP/1.1" 200 OK

2026-05-18 00:44:07,731 [INFO] ehs_api - Step 1: Running CV analysis...

2026-05-18 00:44:08,784 [INFO] ehs_api - Step 2: Querying knowledge base...

Batches:

100% |

1/1 [00:00<00:00, 90.87it/s]

2026-05-18 00:44:08,804 [INFO] ehs_api - Step 3: Running compliance assessment...

2026-05-18 00:44:09,048 [INFO] reasoning agent - Groq client initialized with model: openai/gpt-oss-120b

2026-05-18 00:44:13,901 [INFO] httpx - HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"

2026-05-18 00:44:13,903 [INFO] ehs_api - Step 4: Generating incident report...

2026-05-18 00:44:16,695 [INFO] httpx - HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"

2026-05-18 00:44:16,698 [INFO] ehs_api - Report saved: reports\INC-202605180044-001.json

INFO: 127.0.0.1:56312 - "POST /analyze/image HTTP/1.1" 200 OK

2026-05-18 00:44:16,762 [INFO] watchfiles.main - 1 change detected

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INFO: 127.0.0.1:56274 - "GET /health HTTP/1.1" 200 OK
2026-05-18 00:43:16,821 [INFO] watchfiles.main - 1 change detected
INFO: 127.0.0.1:56305 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:56310 - "GET /health HTTP/1.1" 200 OK
2026-05-18 00:44:07,731 [INFO] ehs_api - Step 1: Running CV analysis...
2026-05-18 00:44:09,784 [INFO] ehs_api - Step 2: Querying knowledge base...
Batches: 100% | 1/1 [00:00:00.00, 90.87it/s]
2026-05-18 00:44:09,884 [INFO] ehs_api - Step 3: Running compliance assessment...
2026-05-18 00:44:09,946 [INFO] reasoning_agent - B B req client initialized with model: openai/gpt-oss-120b
2026-05-18 00:44:13,901 [INFO] httpx - HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
2026-05-18 00:44:13,903 [INFO] ehs_api - Step 4: Generating incident report...
2026-05-18 00:44:16,695 [INFO] httpx - HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
2026-05-18 00:44:16,698 [INFO] ehs_api - Report saved: reports/INC-202605180044-001.json
2026-05-18 00:44:16,762 [INFO] analysis - POST /analyze/image HTTP/1.1" 200 OK
2026-05-18 00:44:16,762 [INFO] watchfiles.main - 1 change detected

```

Detected Voilations:



Detected Violations	
[V001] PPE_MISSING -- HIGH Description: Hard hat not detected on any of the three workers; required per ANSI Z89.1 in areas with overhead hazards or active construction. Policy Reference: PPE-001 Evidence: ppe_details.ppe_status[.hard_hat_detected = false for all three persons	HIGH IMMEDIATE ACTION
[V002] PPE_MISSING -- HIGH Description: Safety footwear (steel-toe) not detected on any worker; mandatory on all production floors (ASTM F2412). Policy Reference: PPE-001 Evidence: ppe_details.ppe_status[.violations includes "safety_shoe" for all three persons	HIGH IMMEDIATE ACTION
[V003] PPE_MISSING -- MEDIUM Description: Safety goggles not detected; required in chemical handling areas and machine shops (ANSI Z87.1). Policy Reference: PPE-001 Evidence: ppe_details.ppe_status[.violations includes "goggles" for all three persons	MEDIUM IMMEDIATE ACTION
[V004] PPE_MISSING -- MEDIUM Description: Chemical-resistant gloves not detected; required when handling acids, bases, or solvents. Policy Reference: PPE-001	MEDIUM IMMEDIATE ACTION

Generating Incident Report:

Deploy

Generated Incident Report

Report ID: [INC-202605180044-001](#)

Location: Industrial Floor A

Date: 2026-05-18 00:44:13

Description: At 00:44:13 on 2026-05-18, the AI monitoring system captured video of three workers operating on Industrial Floor A. Computer-vision analysis confirmed that none of the workers were wearing any of the required personal protective equipment. Specifically, hard hats (ANSI Z89.1 Class E), steel-toe safety footwear (ASTM F2412), safety goggles (ANSI Z87.1), chemical-resistant gloves, and respiratory protection (N95 or higher) were all absent. The area contains overhead hazards and active chemical handling, making the lack of PPE a direct breach of PPE-001 policy and a high-risk safety condition.

Corrective Actions:

[IMMEDIATE]

Immediately stop work in the affected area and require all workers to don the required PPE before resuming operations.

Shift Supervisor

1 hour

[IMMEDIATE]

Document the violation in the incident log and attach supporting AI video evidence.

EHS Coordinator

2 hours

[SHORT_TERM]

Conduct a toolbox talk with the three workers and the entire shift on PPE requirements and consequences of non-compliance.

Safety Trainer

24 hours

[SHORT_TERM]

Review and update PPE training records for the involved personnel; schedule refresher training if gaps are identified.

Training Manager

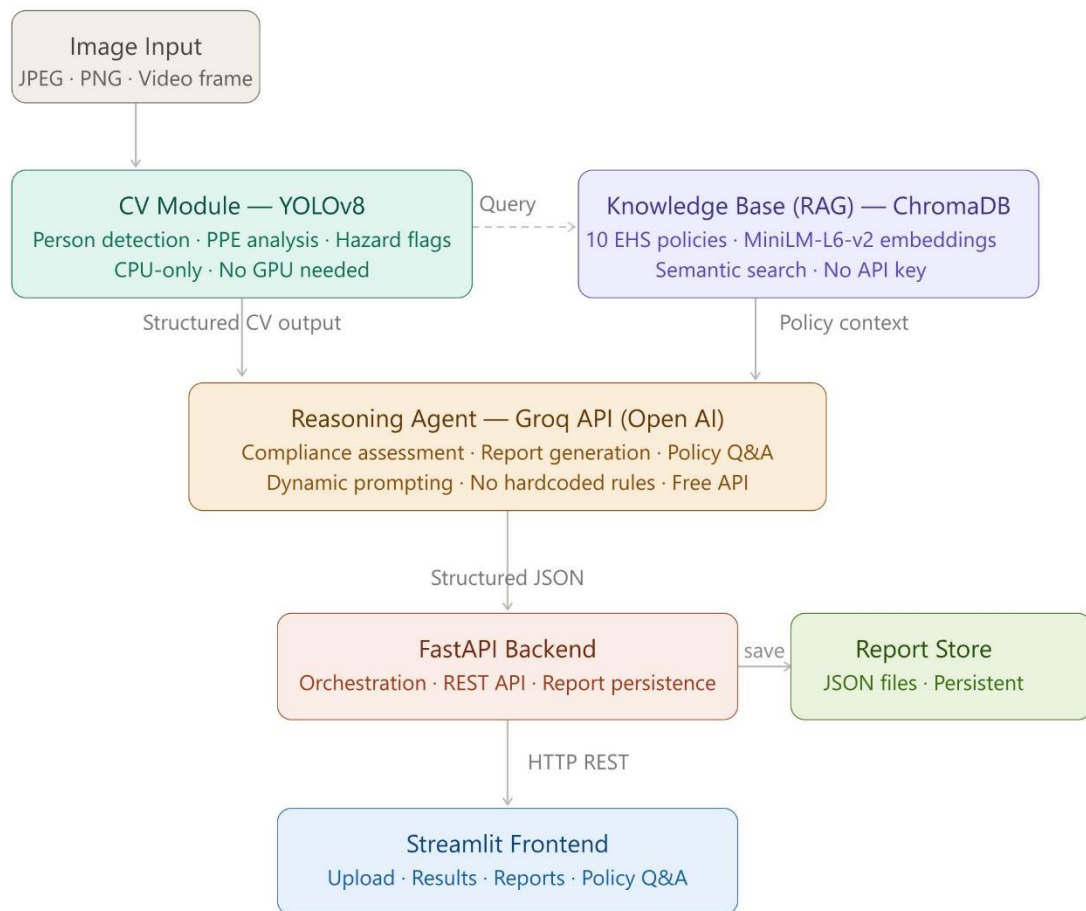
1 week

[SHORT_TERM]

Verify PPE inventory levels on Industrial Floor A and replenish any missing items to ensure immediate availability.

Workflow Architecture:

EHS AI POC — System Architecture



Models used: ■ YOLOv8 (6-12MB) ■ MiniLM-L6-v2 (~22MB) ■ openai/gpt-oss-120b(Groq API)
All CV runs locally on CPU — no cloud needed for detection